PAPIT: Prompt-Aware Pruning for Image Tokens in Efficient Multimodal Inference

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Abstract

*We present **PAPIT-Distill**, a* 1.3*M-parameter MLP that distils LLaVA-1.5-7B's mid-layer attention into a single-pass image-token pruner. On TextVQA at* k=25% *retention (a* 10× *attention-FLOP reduction) it gains* +12 *points over random pruning (*36.0 *vs.* 23.7*), essentially recovering the unpruned baseline (*36.7*). On GQA and VQA v2 at* k=50% *it closes the gap to unpruned to within a single point. Scoring a* 576*-patch image takes* 0.43 *ms —* ∼150× *faster than its CLIP-GradCAM analogue and* ∼500× *faster than the two-pass oracle it learns from. En route we report a quantitative* negative result *that motivates the method: cross-dataset Spearman* $\rho \approx 0$ *between CLIP-derived saliency and LLaVA's own LLM attention, ruling out training-free CLIP-aligned scorers as a class.*

1. Introduction

LLaVA-1.5 [1] composes a CLIP ViT-L/14 vision encoder [2, 3] with a 7B-parameter language model. Each of the $24 \times 24 = 576$ image tokens participates in every self-attention layer [4], making inference cost quadratic in the number of visual tokens. For a typical VQA query fewer than half of these tokens are relevant.

A common intuition is that the user's prompt should be enough to choose which tokens to keep, *without* any additional training. We make this claim concrete: *can a training-free saliency signal, read from LLaVA's own CLIP-initialised vision tower, predict which patches the LLM relies on, well enough to beat a query-agnostic random baseline?*

We answer this in three stages. First (§3) we instantiate the idea as **PAPIT**, a pipeline that inserts a GradCAM-based top-k between the ViT and the MLP projector. Second (§5.1) we show that PAPIT *at best* matches random pruning, and a direct alignment diagnostic (§5.2) explains why: the CLIP-aligned scoring signal has $\rho \approx 0$ with the LLM's own attention. Third (§5.4) we run a multi-layer LLM-attention oracle as a positive control — mid-layer attention *does* beat random — and distil that signal into a 1.3M-param MLP that runs in single-pass, training-free deployment.

Contributions. (i) A clean training-free token-pruning pipeline for LLaVA-1.5 with 41–90% attention-FLOP reduction at $k \in [25\%, 75\%]$. **(ii)** A cross-dataset *negative result*: CLIP-derived patch saliency is statistically uncorrelated with LLaVA's LLM attention, ruling out an entire family of training-free prompt-aware methods. **(iii) PAPIT-Distill**, a small MLP distilled from mid-layer LLM attention that turns the negative result into a deployable single-pass scorer. Code at `github.com/yarikama/PAPIT`.

2. Related Work

ViT-side pruning — DynamicViT [5] and AdaViT [6] learn per-block scorers; ToMe [7] merges similar tokens without retraining. All target classification ViTs and ignore the prompt. **LLM-side pruning for VLMs** — FastV [8] drops low-attention tokens at LLM layer 2; SparseVLM [9] extends this with diversity heuristics. Both need a full forward pass before pruning. **CLIP-aligned signals** — MaskCLIP [10] unlocks dense patch features; register tokens [11] complicate naive top-k. PAPIT differs by pruning *before* the LLM with a query-conditioned score; §5.2 shows the naive CLIP version fails, and §5.4 gives the distillation that fixes it.

3. Method

3.1. Pipeline

LLaVA's frozen CLIP ViT-L/14 produces 576 patch features $H \in \mathbb{R}^{576 \times 1024}$; the prompt is encoded by the matching CLIP text encoder into a single $T \in \mathbb{R}^{768}$. A scoring step returns $s \in \mathbb{R}^{576}$; the top-k patches plus a single *anchor* token (mean of all 576 patches) are projected by LLaVA's MLP and spliced into the LLM input. Position IDs are remapped to a dense $0, \ldots, k$ sequence. We study two scorers below: a training-free GradCAM on LLaVA's ViT (PAPIT-CLIP) and a 1.3M-param MLP distilled from mid-layer LLM attention

(PAPIT-Distill).

3.2. Scorer 1: PAPIT-CLIP (GradCAM)

CLIP's CLS-only contrastive objective makes naive patch–text cosine unstable, so we use Grad-CAM [12]: with $h_i^{(L-1)}$ the penultimate-ViT activation of patch i and $\mathrm{sim}(\mathrm{CLS}, T)$ the projected CLS–text cosine, set

$$s_i^{\mathrm{gc}} = \mathrm{ReLU} \sum_d \frac{\partial\, \mathrm{sim}}{\partial\, h_{i,d}^{(L-1)}} \cdot h_{i,d}^{(L-1)} \tag{1}$$

and take top-k. Section 5.2 will show this signal is essentially uncorrelated with what the LLM actually attends to.

3.3. Scorer 2: PAPIT-Distill

A multi-layer two-pass oracle (Table 3) confirms the missing signal lives *inside* the LLM, with mid-layer attention ($L \in [8, 24]$) beating random. PAPIT-Distill mimics this at single-pass cost. The predictor is a 1.3M-param MLP_4: $z_i^{(0)} = [W_p h_i \| W_t T] \in \mathbb{R}^{2d}$ with d=384, passed through three LayerNorm–Linear–GELU blocks with residual shortcut, $z^{(\ell)} = z^{(\ell-1)} + f_\ell(z^{(\ell-1)})$ for ℓ=1,2,3, and read out by a linear head, giving $s_i^{\mathrm{d}} = w^\top z_i^{(3)}$. We supervise against the layer-8 attention $A^{(8)}$ summed over non-image query rows and head-averaged into a distribution y over the 576 image-token positions, with KL loss

$$\mathscr{L} = \mathrm{KL}\big(y \,\|\, \mathrm{softmax}(s^{\mathrm{d}})\big), \tag{2}$$

on a 100K-pair mix (50K GQA + 30K VQAv2 + 20K TextVQA). $L = 8$ is selected by ablation across $L \in \{2, 8, 16, 24, 31\}$ plus attention rollout: distillability beats oracle strength — L=8 predictor gains +0.030 over random vs. +0.024 for L=16 and −0.054 (worse than random) for rollout.

3.4. Efficiency

Self-attention cost is quadratic in sequence length. Let k be the number of retained image tokens and $L_{\mathrm{text}} \approx 50$ the prompt length; then

$$\mathrm{FLOPs}_{\mathrm{rel}} = \frac{(k + L_{\mathrm{text}})^2}{(n + L_{\mathrm{text}})^2}, \tag{3}$$

giving a 10× attention-FLOP reduction at k=25% (n=576). **Wall-clock scoring latency** on g5.2xlarge (A10G, 20 samples, mean ± std, retention 0.5): random 0.07±0.00 ms, PAPIT-Distill **0.43±0.00** ms, PAPIT-CLIP (GradCAM) 65.23±6.08 ms, two-pass Oracle L=16 216.17±1.17 ms. PAPIT-Distill is therefore ∼150× faster than PAPIT-CLIP and ∼500× faster than the oracle, while

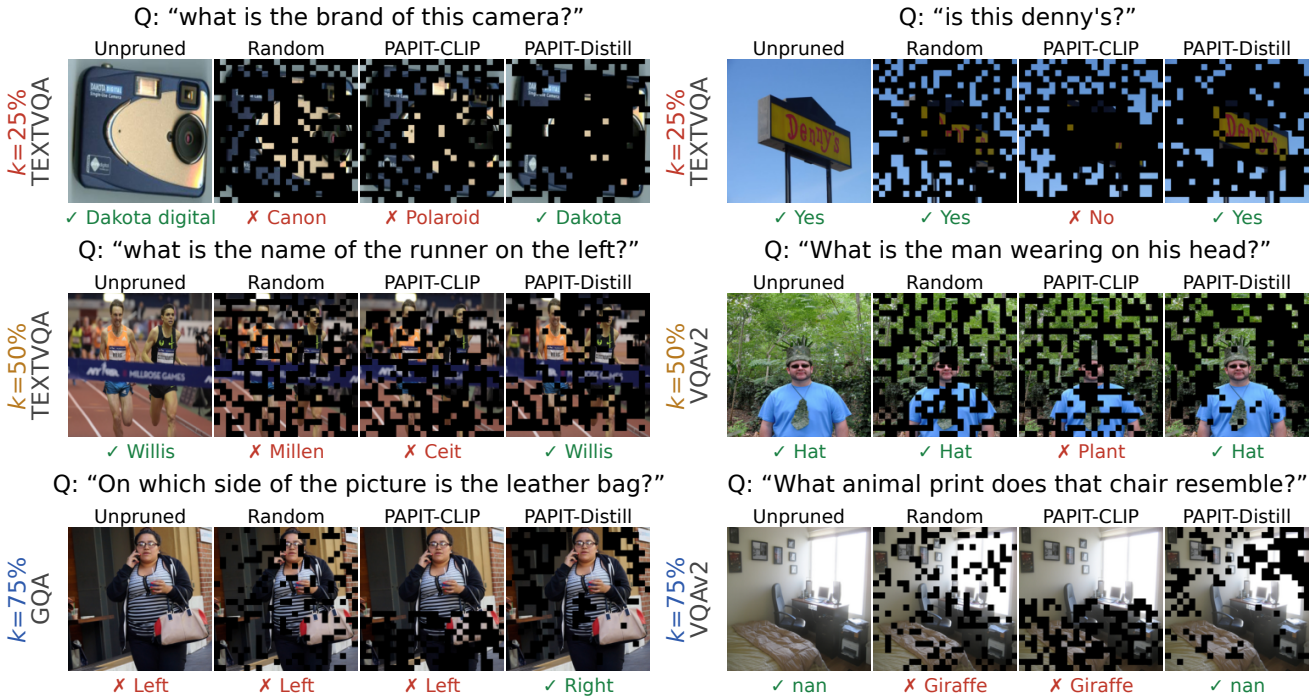

Figure 1. **Real input, real output across three benchmarks and three retention ratios.** Six cases where PAPIT-Distill answers correctly while PAPIT-CLIP fails, drawn from TextVQA / GQA / VQAv2 at $k \in \{25\%, 50\%, 75\%\}$ (colour-coded). Each row shows the question; columns are Unpruned / Random / PAPIT-CLIP / PAPIT-Distill. Black = pruned patches; the green check / red cross under each panel marks whether LLaVA's actual output matches the gold answer.

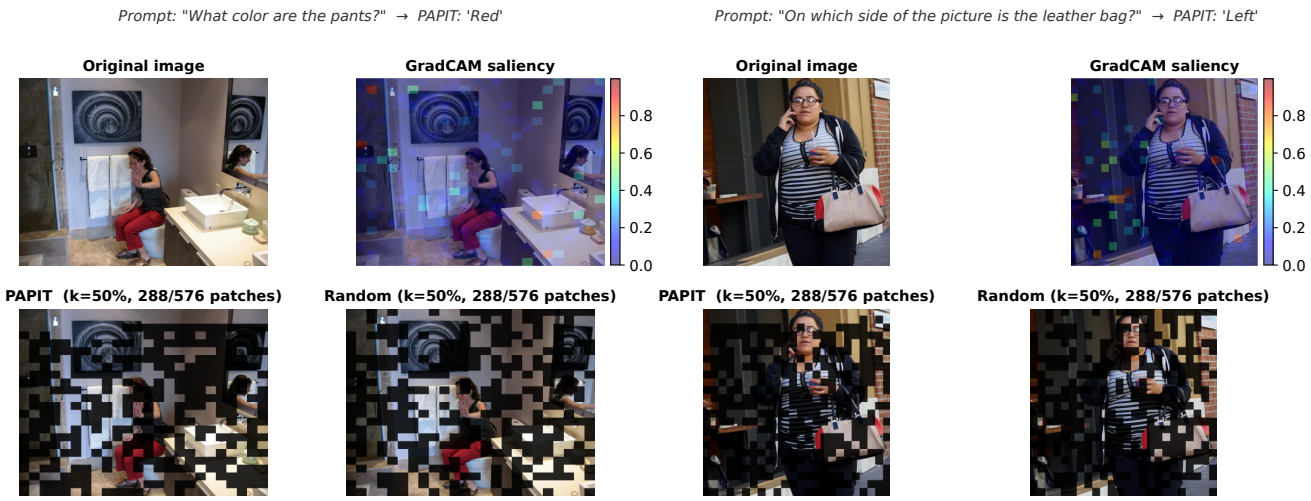

Figure 2. **PAPIT-CLIP localises but does not align** (k=50%, GQA). Saliency concentrates on the query-relevant region, yet Sec. 5.2 shows it is *statistically uncorrelated* with what LLaVA's LLM attends to.

delivering an accuracy gain comparable to the oracle (Table 1).

4. Experiments

Models. LLaVA-1.5-7B (`llava-hf/llava-1.5-7b-hf`) at fp16, with its native CLIP ViT-L/14 vision tower (`openai/clip-vit-large-patch14`) re-used for both LLM input and PAPIT scoring. PAPIT-Distill predictor: 1.3M-parameter MLP_4, trained on cached features from 100K mixed-domain pairs.

Datasets. GQA [13] test-dev balanced, VQA v2 [14] val-balanced, and TextVQA [15] val. Each held-out split uses 100 samples per (dataset, retention) cell for downstream accuracy and 100 samples for diagnostics; binomial standard error at N=100 is ≈ 0.05. Accuracy: exact match for GQA, VQA soft accuracy $\min(\text{count}/3, 1)$ [14] for the others.

Baselines. **Unpruned** keeps all 576 tokens. **Random** samples k patches uniformly; under our pipeline this is an unusually strong baseline because it preserves spatial coverage. **Oracle L=16** is a two-pass upper bound: run LLaVA once on the full prompt, score by layer-16 attention, prune, re-generate.

Table 1. **Main accuracy** on LLaVA-1.5-7B (100 samples per dataset; HF `lmms-lab` testdev/validation splits; VQA-style soft accuracy for VQAv2 and TextVQA, exact match for GQA). *Higher is better.* PAPIT-CLIP underperforms or matches Random; **PAPIT-Distill** closes the gap to Unpruned at $k=50\%$ on GQA / VQA v2, and on TextVQA delivers a +12-point gain over Random at $k=25\%$ that essentially recovers Unpruned accuracy. Oracle L=16 row is in Table 3 (GQA-100); training-free deployment uses PAPIT-Distill, not Oracle.

Dataset	Method	k=25%	k=50%	k=75%
Rel. FLOPs (Eq. 3)		*0.10*	*0.29*	*0.59*
GQA	Unpruned		57.0	
	Random	52.0	52.0	55.0
	PAPIT-CLIP	55.0	52.0	50.0
	PAPIT-Distill	53.0	**56.0**	55.0
VQA v2	Unpruned		83.3	
	Random	81.7	81.0	82.0
	PAPIT-CLIP	72.3	80.3	82.7
	PAPIT-Distill	79.3	**83.3**	**85.0**
TextVQA	Unpruned		36.7	
	Random	23.7	30.7	32.0
	PAPIT-CLIP	20.0	28.3	31.7
	PAPIT-Distill	**36.0**	**36.0**	**33.3**

5. Results

5.1. PAPIT-CLIP matches but does not exceed random

Table 1 summarises the headline accuracy. The GradCAM-based PAPIT-CLIP scorer is at-best level with Random on most (dataset, retention) cells (gaps $<1\sigma$, with $\sigma \approx 0.05$ at N=100), and notably *worse* on VQA v2 $k=25\%$ (a 9.4-point gap, $\sim 1.9\sigma$ below Random). Figures 1 and 2 show PAPIT-CLIP *does* produce visibly localised selections, yet this does not translate to a measurable downstream gain.

5.2. Why CLIP saliency is uninformative

We compare s^{gc} to the LLM's own attention $s^{\text{llm}} \in \mathbb{R}^{576}$ (last-layer, head-averaged, summed across the first 8 generated answer tokens) per image. A uniform-random selector yields top-10% Jaccard ≈ 0.053.

The two signals are essentially uncorrelated on every

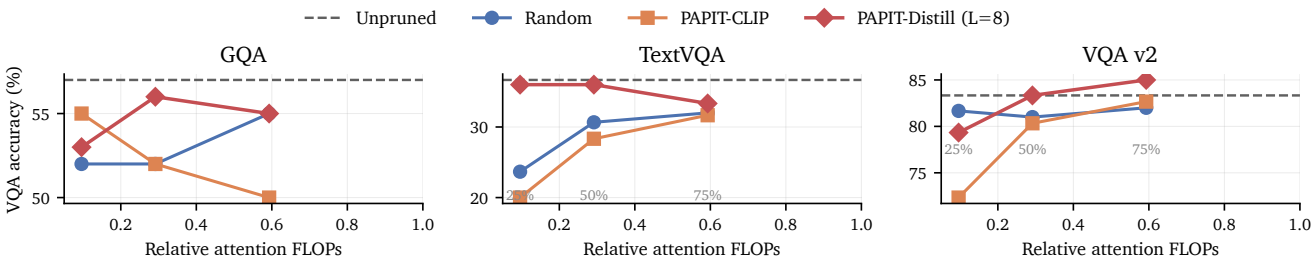

Figure 3. **Accuracy–FLOPs Pareto on LLaVA-1.5-7B**, $N=100$ per dataset. PAPIT-Distill (red diamond) is on or above the Random/PAPIT-CLIP curve on 7 of 9 (dataset, retention) cells; at TextVQA $k=25\%$ ($\sim 10\times$ FLOP reduction) it gains +12.3 over random and essentially recovers Unpruned. Annotations on each x-tick give the retention k.

Table 2. **Alignment diagnostic** between PAPIT-CLIP scores and LLaVA's last-layer LLM attention over 576 image tokens (100 samples per dataset). All three datasets give $\rho \approx 0$ and top-10% Jaccard *below* the uniform-random reference of 0.053, so any CLIP-aligned scorer is at-best random downstream.

Metric	GQA	TextVQA	VQA v2
Spearman ρ mean	−0.018	−0.016	0.014
Spearman ρ median	0.008	0.019	0.010
% of samples $\rho > 0.3$	0.0%	0.0%	0.0%
% of samples $\rho < 0$	48%	47%	43%
Top-10% Jaccard	0.047	0.039	0.046

benchmark, and PAPIT-CLIP's top-10% patches overlap with the LLM's most-attended patches less than uniform-random selection would. This rules out an entire family of training-free pruners that score patches in any CLIP-aligned space, regardless of the specific scoring rule (see [10] for the dense-feature analogue).

5.3. Mid-layer LLM attention does carry the signal

We test the converse with a two-pass oracle: run LLaVA once on the full prompt, score by attention from non-image queries to image keys at LLM layer L, prune, and re-generate. Table 3 sweeps $L \in \{2, 8, 16, 24, 31\}$ on GQA. Mid-layer attention ($L \in [8, 24]$) consistently beats random; layer 2 (FastV's recommended layer in its single-pass setting) and the last layer do not. This is the positive control that motivates distillation.

5.4. PAPIT-Distill: making the oracle deployable

The oracle costs a second LLaVA forward; we distil it. An architecture sweep at supervision target $L=16$ (Bilinear / MLP_2 / MLP_4 / cross-attn, 5K-sample held-out) selects MLP_4 as the lightest predictor reaching val-KL 0.087 and top-10% Jaccard 0.62. Downstream accuracy follows *distillability*, not oracle strength: $L=8$ predictor gains +0.030 over random vs. +0.024 for $L=16$, +0.021 for $L=24$, and −0.054 for attention rollout [16] — the rollout target aggregates all 32 layers and is correspondingly

Table 3. **Two-pass LLM-attention oracle on GQA** (100 samples, exact match). **Bold**: beats random at that retention. Layer 16 gives the largest margin at low retention; FastV's layer 2 fails in this two-pass setup because the prompt-only forward has not yet committed attention mass to query-relevant patches.

Selector	k=25%	k=50%	k=75%
Unpruned	0.58	0.58	0.58
Random	0.53	0.53	0.56
PAPIT-CLIP	0.56	0.53	0.51
LLM-attn $L=2$	0.49	0.54	0.53
LLM-attn $L=8$	**0.58**	**0.54**	0.56
LLM-attn $L=16$	**0.61**	**0.58**	0.56
LLM-attn $L=24$	**0.60**	**0.57**	0.55
LLM-attn $L=31$	0.50	**0.55**	0.55

noisy. The deployed predictor is MLP_4 on $A^{(8)}$, visualised in Figure 3.

6. Discussion and Conclusion

Why is PAPIT-CLIP rarely worse than random? LLaVA-1.5 has heavy token redundancy [8], so at $k \geq 50\%$ the accuracy driver is the *number* of kept tokens, not their identity. The exception is VQA v2 $k=25\%$, where dropping the wrong 75% costs $\sim$9 points. Layer 2 attention fails for a related reason: the prompt-only forward has not yet committed attention to query-relevant patches and is dominated by register-like positions [11]; layers 8–24 have integrated question with image; $L=31$ has already shifted to generation.

Limitations and outlook. All numbers are at $N=100$ per (dataset, retention) cell; binomial $\sigma \approx 0.05$, so the TextVQA $k=25\%$ gain (+12.3) is $\sim 2.5\sigma$ — worth replicating at $N=500+$. Extending to LLaVA-1.6, Qwen2.5-VL, and FastV-style single-pass integration are natural next steps.

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